

Summary Measures Worksheet – Research Version

Exercise 6.1 and 6.2 – Diet A and Diet B

Table of summary measures:

Measure	Diet A	Diet B
Sample size (n)	50	50
Mean weight loss	5.341 kg	3.710 kg
Standard deviation	2.536 kg	2.769 kg
Median weight loss	5.642 kg	3.745 kg
Interquartile range (IQR)	3.285 kg	3.812 kg

Interpretation

Central tendency

- Both the mean and median weight loss are higher for Diet A than for Diet B.
- On average, participants on Diet A lost around 1.6 kg more than those on Diet B.

Variability

- Diet A has a slightly smaller standard deviation, so weight loss outcomes are a little more clustered around the mean.
- The IQR is marginally larger for Diet B, which indicates more spread among the middle 50% of weight-loss values.

Comparative effectiveness

- Diet A appears more effective overall, because it combines higher typical weight loss (mean and median) with similar or slightly lower variability than Diet B.
- In practical terms, someone choosing between the two diets would expect larger and more consistent weight loss on Diet A.

Exercise 6.3 – Brand Preferences in Two Areas

From the worksheet for Area 1:

- Brand A: 15.7%
- Brand B: 24.3%
- Other brands: 60.0%

From the worksheet for Area 2:

- Brand A: 21.1%
- Brand B: 33.3%
- Other brands: 45.6%

Interpretation

- Area 2 shows a stronger preference for named brands (A and B) than Area 1.
- Area 1 respondents are more inclined to choose “Other” brands, suggesting a less concentrated pattern of loyalty.
- For marketing strategy, this could mean:
 - In Area 2, reinforcing brand identity and promotions for A and B may be productive, because consumers already lean toward these brands.
 - In Area 1, there may be space either to introduce new brands or to reposition existing brands, as a large proportion of customers look beyond the main labelled options.

References

Field, A. (2018) Discovering statistics using IBM SPSS Statistics. 5th edn. London: SAGE.

Newbold, P., Carlson, W. and Thorne, B. (2013) Statistics for business and economics. 8th edn. Harlow: Pearson.